

Hanyu Chen

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EDUCATION

Cornell University Ph.D. in Computer Science Advisor: Prof. Noah Snavely	Aug 2024 – Present
Carnegie Mellon University M.S. in Computer Science (QPA 4.08/4.3) Advisor: Prof. Ioannis Gkioulekas	Jun 2023 – Jun 2024
B.S. in Computer Science (QPA 3.94/4.0) Additional major in Mathematics; minor in Computer Graphics	Aug 2019 – Jun 2023

RESEARCH INTERESTS

3D reconstruction, neural rendering, computer graphics, 3D computer vision

PUBLICATIONS

- [1] **Honey, I Shrunk the Arc de Triomphe!** ([project link](#))
Yuanbo Xiangli, **Hanyu Chen**, Xueqing Tsang, Noah Snavely
European Conference on Computer Vision, 2026.
- [2] **ArchSym: Detecting 3D-Grounded Architectural Symmetries in the Wild** ([project link](#))
Hanyu Chen, Ruojin Cai, Steve Marschner, Noah Snavely
IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.
- [3] **HairFormer: Transformer-Based Dynamic Neural Hair Simulation** ([arXiv](#))
Joy Xiaoji Zhang, Jingsen Zhu, **Hanyu Chen**, Steve Marschner
arXiv, 2025.
- [4] **Doppelgangers++: Improved Visual Disambiguation with Geometric 3D Features** ([project link](#))
Yuanbo Xiangli, Ruojin Cai, **Hanyu Chen**, Jeffrey Byrne, and Noah Snavely
IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025. **Highlight.**
- [5] **3D reconstruction with fast dipole sums** ([project link](#))
Hanyu Chen, Bailey Miller, and Ioannis Gkioulekas
ACM Transactions on Graphics (SIGGRAPH Asia), 2024
- [6] **Objects as volumes: A stochastic geometry view of opaque solids** ([project link](#))
Bailey Miller, **Hanyu Chen**, Alice Lai, and Ioannis Gkioulekas
IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024. **Best Student Paper Honorable Mention.**

JOB EXPERIENCE

Software Engineer Intern Map Engine Team, Nvidia	Summer 2022 Santa Clara, CA (Remote)
Curated dataset of ~200k road images with misalignment and calibration issues from map review issue logs. Trained an image classification model to detect invalid road images with a high recall rate, reducing workload for manual review.	
Algorithm Engineer Intern WLAN Team, Huawei	Summer 2021 Beijing, China
Simulated a 5 GHz wireless network with ~150 access points in an office environment. Implemented bipartite matching and greedy depth-first search algorithms for dynamic channel allocation, minimizing co-channel interference.	

PROFESSIONAL SERVICE

Teaching Assistant

- Data Structures and Functional Programming
- Computer Graphics Practicum
- Physics-based Rendering
- Algebraic Structures

Spring 2025 · Cornell
Fall 2024 · Cornell
Spring 2023 · CMU
Fall 2022 · CMU

Reviewer: CVPR, ECCV, SIGGRAPH Asia, NeurIPS

ACADEMIC PROJECTS

Adaptive LiDAR sampling based on free-flight uncertainty

Fall 2023 · CMU

Computational Photography

- Developed a novel adaptive LiDAR sampling scheme for scanning objects by progressively placing samples at locations of high uncertainty, characterized by the entropy of the free-flight distribution of randomly sampled rays.

Differentiable rendering for optimizing local scene parameters

Spring 2022 · CMU

Physics-based Rendering

- Implemented a path-tracing-based renderer in C++ to compute gradients of a rendered image with respect to local scene parameters. Optimized material properties and lighting to match target images using gradient descent.

CUDA-Based bag-of-words scene recognition

Spring 2022 · CMU

Parallel Computer Architecture and Programming

- Parallelized convolution, extraction, and clustering stages of a bag-of-words classifier in C++ and CUDA, resulting in a 50x speedup over a sequential algorithm, and an 8x speedup over a baseline OpenMP implementation.

Animating hand-drawn sketches using image autoencoders

Fall 2021 · CMU

Visual Learning and Recognition

- Implemented a CNN-based autoencoder with an auxiliary discriminator network to animate hand-drawn sketches by interpolating between latent vectors and reconstructing keyframes.

SKILLS

Languages: C, C++, Python, OCaml, Standard ML

Frameworks/Libraries: PyTorch, OpenCV, Open3D, NumPy, Eigen, CUDA, OpenMP, Git